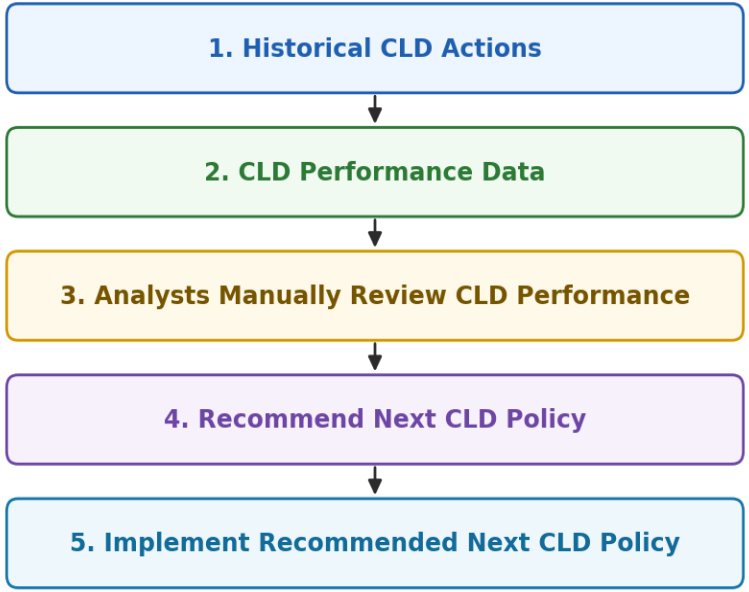
Offline Reinforcement Learning for Credit Line Decrease (CLD) Policy Optimization

A Simulated Framework for Segment-Level Policy Recommendation, Business Impact Evaluation, and Continuous Policy Improvement

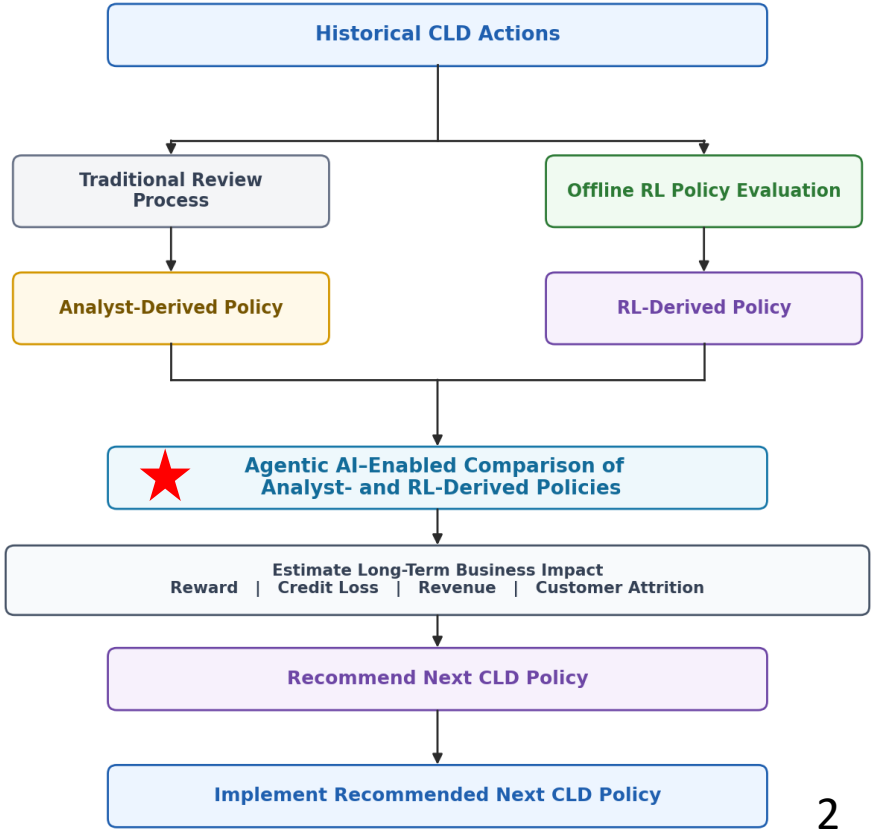
Jingru Chen | 2026-07-18

Executive Summary: Enhancing CLD Policy Evaluation with Offline RL

Traditional CLD Review Process



Offline RL Enhanced CLD Process



Case Study: Simulated Offline RL Framework for CLD Policy Optimization

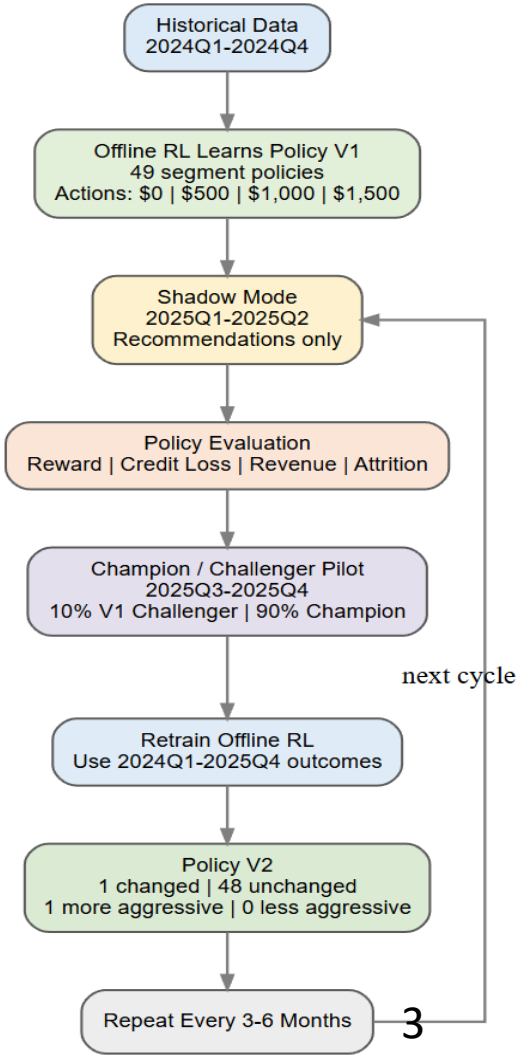
Simulated Offline RL Framework:

(1) Historical Data
Simulated quarterly CLD portfolio (2024Q1–2025Q4)
49 customer segments
Account-level behavioral outcomes

(2) Behavior Models – **We can use bank-already-deployed models**

(3) Offline RL
Learn optimal CLD policy by segment
Compare with existing policy
Estimate long-term business impact

(4) Outputs
Estimated business impact (reward, credit loss, revenue, attrition)
Segment-level Policy V1
Segment-level Policy V2
Policy comparison dashboard
Champion/Challenger summary



Sample Offline Recommended CLD Policies

Segment-Level CLD Policy Matrix

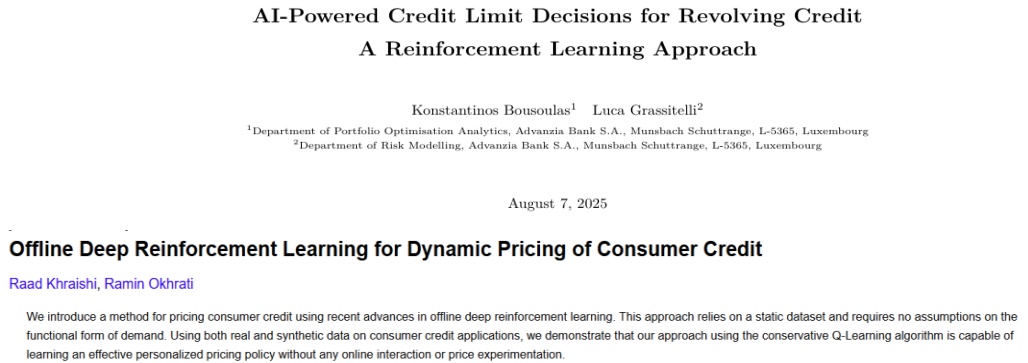
Fixed-dollar Policy V1 versus Policy V2

	Segment Definition			Policy Decision					Evidence & Diagnostics		
Segment ID	FICO	Utilization	Delinquency	V1 CLD	V2 CLD	V1 → V2	Status	Direction	Est. Reward Lift	Pilot Effect	Decision Reason
600-659|60%-84%|Current	600-659	60%-84%	Current	No CLD	$500	**No CLD → $500**	Changed	More aggressive	194.96	−36.49	Select highest eligible adjusted Q reward
600-659|30%-59%|Current	600-659	30%-59%	Current	No CLD	$500	**No CLD → $500**	Changed	More aggressive	157.43	−2.59	Select highest eligible adjusted Q reward
<600|30%-59%|Current	<600	30%-59%	Current	No CLD	$500	**No CLD → $500**	Changed	More aggressive	143.99	5.56	Select highest eligible adjusted Q reward
660-719|60%-84%|Current	660-719	60%-84%	Current	No CLD	$500	**No CLD → $500**	Changed	More aggressive	125.63	21.48	Select highest eligible adjusted Q reward
600-659|30%-59%|1-29 DPD	600-659	30%-59%	1-29 DPD	No CLD	No CLD	**No CLD → No CLD**	Unchanged	No change	0.00	11.92	V1 remains highest eligible action
600-659|30%-59%|30-59 DPD	600-659	30%-59%	30-59 DPD	$1,500	$1,500	**$1,500 → $1,500**	Unchanged	No change	0.00	0.00	V1 remains highest eligible action
600-659|30%-59%|60+ DPD	600-659	30%-59%	60+ DPD	$1,500	$1,500	**$1,500 → $1,500**	Unchanged	No change	0.00	0.00	V1 remains highest eligible action
600-659|60%-84%|1-29 DPD	600-659	60%-84%	1-29 DPD	No CLD	No CLD	**No CLD → No CLD**	Unchanged	No change	0.00	5.56	V1 remains highest eligible action
600-659|60%-84%|60+ DPD	600-659	60%-84%	60+ DPD	$1,500	$1,500	**$1,500 → $1,500**	Unchanged	No change	0.00	5.56	V1 remains highest eligible action
600-659|85%+|1-29 DPD	600-659	85%+	1-29 DPD	No CLD	No CLD	**No CLD → No CLD**	Unchanged	No change	0.00	5.56	V1 remains highest eligible action
600-659|85%+|Current	600-659	85%+	Current	No CLD	No CLD	**No CLD → No CLD**	Unchanged	No change	0.00	0.00	V1 remains highest eligible action

... ...

Industry Adoption of Offline Reinforcement Learning

- ✓ Several leading financial institutions and consulting firms are investing in reinforcement learning research and applications for sequential financial decision-making.
- ✓ My CLD demonstration is an example of how these ideas could be adapted to credit line policy optimization.



AI-Powered Credit Limit Decisions for Revolving Credit
A Reinforcement Learning Approach

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August 7, 2025

Offline Deep Reinforcement Learning for Dynamic Pricing of Consumer Credit

Raad Khraishi, Ramin Okhrati

We introduce a method for pricing consumer credit using recent advances in offline deep reinforcement learning. This approach relies on a static dataset and requires no assumptions on the functional form of demand. Using both real and synthetic data on consumer credit applications, we demonstrate that our approach using the conservative Q-Learning algorithm is capable of learning an effective personalized pricing policy without any online interaction or price experimentation.

https://www.jpmorganchase.com/about/technology/research/machine-learning
https://capitalone.science/research/
https://www.citi.com/ventures/perspectives/opinion/agents-as-a-service-evolution.html
https://www.ey.com/content/dam/ey-unified-site/ey-com/en-ca/services/ai/documents/ey-enhancing-financial-risk-management-with-reinforcement-learning.pdf

Beyond CLD: A Reinforcement Learning Framework for Credit Line MGT

Potential Offline Reinforcement Learning Applications

Credit Line Assignment (CLA): Optimize initial credit limits by balancing revenue and credit risk.

Credit Line Increase (CLI): Recommend optimal line increases using long-term customer behavior.

Credit Line Decrease (CLD): Optimize CLD policies using historical portfolio outcomes.

Portfolio Monitoring: Monitor policy performance and detect deterioration.

Policy Evaluation: Shadow Mode and Champion/Challenger policy evaluation.

Policy Optimization: Retrain Offline RL every 3–6 months to recommend the next policy version.

Key Message

Offline Reinforcement Learning complements existing credit risk models by learning from historical portfolio outcomes to improve CLA, CLI, CLD, policy evaluation, and continuous policy optimization.

Thank You

Questions & Discussion